



Faculty Of Information Science & Technology

TDM6323– Data Mining and Machine Learning

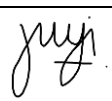
Trimester 3 – 2024 / 2025

Assignment Report

Title: Rental Pricing Dataset

Lab Session 1A

Group Number 2

	Student ID	Student Name	Programme	Contribution Percentage	Signature
1	1191202283	Uithiswary a/p Velan	BIA	25%	
2	1211203835	Liew Zhi Tong	BIA	25%	
3	1211204549	Priveeta Gobi Krishnan	BIA	25%	
4	1211201569	Yeo Hui Yi	BIA	25%	
*Programme: AI / BIA / DCN / ITM / ST/BI					

Project Declaration Form

We hereby declare that all group members' names are correctly included in the section above and all members have contributed to this project. We hold a copy of this assignment, which we can produce if the original is lost or damaged. We certify that no part of this assignment has been copied from any other student's work or any other source except where due acknowledgement is made in the report.

We also acknowledge that the use of AI tools (if any) is aimed at brainstorming and enhancing decision-making processes. We affirm our commitment to ethical guidelines, transparency, and responsible utilization of AI technology in acquiring solutions.

	Marks	
1 Introduction		5
2 Dataset Selection, Description		5
3 Data Cleaning		5
4 Descriptive Statistics		5
5 Data Visualization		5
6 Insights and Interpretation		15
7 Conclusion		5
8 Reference & Format		5
9 Programming		10
Total Marks		60
Final Marks		30

Group Member's Contribution to the Task

(Each group member, including the group leader, must individually fill out and submit this form. Elaborate on the tasks and contributions in this project. This form must be attached to the project submission.)

I, **Uithiswary a/p Velan** hereby declare that I have assessed the final submission, and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays, or non-submission. For the purpose of completing this project, we have worked as a team, and I have mainly contributed to the following tasks:

1. I distributed the main task for each member for this project.
2. From the distribution, I had done and lead the other team members on Data Visualization part.
3. I followed all the instruction and contributed in the part's other members are leading which are introduction, data cleaning, insights/ interpretation and conclusion.

I, **LIEW ZHI TONG** hereby declare that I have assessed the final submission and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays, or non-submission. For the purpose of completing this project, we have worked as a team and I have mainly contributed in the following tasks:

1. I contributed as a member of the team and was responsible for the Introduction and Conclusion sections of the report.
2. I followed all the instructions and actively contributed to the sections led by other team members, including Data Visualization, Data Cleaning, and Insights/Interpretation.

I, **PRIVEETA GOBI KRISHNAN** hereby declare that I have assessed the final submission, and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays, or non-submission. For the purpose of completing this project, we have worked as a team, and I have mainly contributed to the following tasks:

1. I was the lead member for the insights and interpretation section of this assignment.
2. I also contributed to all the sections led by other team members, including Introduction, Data cleaning and Description, Data Visualization and Conclusion.

I, **YEO HUI YI** hereby declare that I have assessed the final submission, and I take full responsibility should there be any inaccuracies, incompleteness, omissions, delays, or non-submission. For the purpose of completing this project, we have worked as a team, and I have mainly contributed to the following tasks:

1. I had done and lead the other team members on Data Selection and Description & Data Cleaning section.
2. I followed all the instruction and contributed in the part's other members are leading which are Introduction, Data Visualization, Insights and Interpretation & Conclusion.

Marking Rubric

	Excellent	Good	Average	Poor	Very Poor
Introduction (5)	5	4	3	2	1-0
	A well-written introduction of the project.	An adequate introduction to project report.	Student has performed moderately with typical results.	Missing section or poorly written introduction.	Student is not able to achieve the task.
Dataset Selection, Description (5)	5	4	3	2	1-0
	An excellent and appropriate dataset choice.	A good and relevant dataset choice.	A moderately appropriate dataset choice.	A less relevant dataset choice.	An inappropriate or no dataset choice.
Data Cleaning (5)	5	4	3	2	1-0
	Thoroughly cleaned and well documented data handling.	Adequately cleaned with minor issues.	Moderately cleaned with some issues.	Poor data cleaning with significant issues.	Very poorly cleaned or not cleaned at all.
Descriptive Statistics (5)	5	4	3	2	1-0
	Comprehensive and accurate summary statistics.	Good summary statistics with minor inaccuracies.	Moderate summary statistics with some inaccuracies.	Poor summary statistics with significant inaccuracies.	Very poor or no summary statistics.
Data Visualization (5)	5	4	3	2	1-0
	Highly effective, varied, and insightful visualizations.	Good visualizations with minor issues.	Moderate visualizations with some issues.	Poor visualizations with significant issues.	Very poor or no visualizations.
Insights and Interpretation (15)	15-13	12-10	9-7	6-4	3-0
	Deep and relevant insights with thorough discussion.	Good insights with adequate discussion.	Moderate insights with some discussion.	Poor insights with minimal discussion.	Very poor or no insights.
Conclusion (5)	5	4	3	2	1-0
	Strong summary of key findings.	A good summary with minor gaps.	Adequate summary with some gaps.	Weak summary with significant gaps.	No conclusion or a very poor summary.
Reference & Format (5)	5	4	3	2	1-0
	Perfectly formatted and well-referenced.	Properly formatted with minor issues.	Acceptable format with some referencing issues.	Poorly formatted with significant referencing issues.	Very poorly formatted or has no references.
Programming (10)	10-9	8-7	6-5	4-3	2-0
	Well-organized, clear, and easy to follow.	Good organization with minor issues.	An acceptable organization with some clarity issues.	Poorly organized with significant clarity issues.	Very poorly organized or unclear.

1.0 Introduction

The rental markets of Kuala Lumpur and Selangor are of considerable importance to Malaysia's cities due to the high demand for accommodation and the significant economic growth of the region. Different rental rates are generated by the various types of properties that are represented in various locations. The rental prices are primarily determined by the type of property, location, and services offered. Despite the increasing prevalence of websites such as mudah.my for searching for rental properties, it can be challenging for landlords and tenants to ascertain whether the price of a particular property is reasonable. A lack of transparency leads to an information vacuum that complicates the process of locating reasonably priced housing for tenants and determining realistic rent rates for landlords. Additional information regarding the market implications of rental price fluctuations and the factors that influence them would facilitate our resolution of this matter.

This project utilizes the Rental Pricing Dataset, Malaysia, which contains detailed information on rental properties in Kuala Lumpur and Selangor. By analysing this dataset using exploratory data analysis (EDA) techniques in Python, particularly with the Pandas library, the study aims to uncover patterns and relationships within the rental market. A variety of property attributes, including type, size, location, and facilities, will be examined to provide insights into the factors influencing rental prices and how these factors vary across different regions.

The primary challenge of this study is identifying the most significant factors that influence rental prices and examining how these factors vary geographically within Kuala Lumpur and Selangor. Listing attributes (e.g., property type, size, and amenities) and location-specific influences (e.g., proximity to public transportation and points of interest) are particularly crucial, as they directly affect a property's perceived value and market competitiveness (Alves, 2024). Understanding these dynamics is essential for enabling data-driven decisions by key stakeholders such as policymakers, landlords, and tenants. To achieve this, the study has four main objectives: first, to clean and analyse the dataset; second, to visualize trends and perform descriptive statistics; third, to identify key factors influencing rental pricing; and finally, to interpret the findings and derive practical, actionable insights.

By analysing this dataset and conducting a rental pattern investigation, the research contributes to the understanding of price dynamics in the rental markets of Kuala Lumpur and Selangor. The findings are expected to provide tenants with practical information to help them identify reasonably priced homes, aid landlords in optimising pricing policies, and guide policymakers in the management of urban housing affordability. The objective of this investigation is to eliminate the knowledge gap and establish a solid foundation for informed rental property market decision-making.

2.0 Data Selection and Description

This dataset consists of property rental listings in Kuala Lumpur and Selangor. It is extremely relevant to the analysis of real estate trends. The dataset contains crucial information that enables an in-depth investigation of the variables affecting rental prices and property preferences. Key characteristics that are essential for assessing the dynamics of the rental market include property type, location, size, monthly rent, and furnishing status. These factors allow for a more thorough understanding of the supply and demand in the market by revealing information about the kinds of property that are available, their prices, and the provided facilities.

There is a variety of property factors in the dataset, such as the number of rooms, bathrooms and parking spots, enhances the analysis by enabling us to investigate the ways in which qualities impact rental desirability. Metrics like completion year, which allow us to evaluate how property age affects rental preferences and pricing.

This dataset is a great choice due to its comprehensive coverage of factors that are crucial for examining the rental market. Its detailed structure guarantees solid analysis and closely matches our

project's goals. This dataset offers the basis for significant insights into market trends and property preferences in Kuala Lumpur and Selangor by concentrating on important elements that influence renting decisions.

Table 2.1 Data Dictionary

Attributes	Description
ads_id	Unique identifier of each listing
prop_name	Property name
completion_year	Completion year of property
monthly_rent	Monthly rental price
location	Location of property
property_type	Type of property
rooms	Number of rooms
parking	Number of parking spaces
bathroom	Number of bathrooms
size	Size of property
furnished	Furnishing status
facilities	Available facilities
additional_facilities	Additional facilities
region	Region of property

3.0 Data Cleaning

We implemented an organized data cleaning techniques to guarantee the accuracy, consistency and reliability of the dataset. Missing values in numeric columns like completion_year, parking, and bathroom were imputed using the median to preserve data consistency and reduce the impact of outliers. While missing values for categorical columns such as prop_name, furnished, and facilities were substituted with the mode to maintain logical consistency throughout the dataset.

Furthermore, formatting errors were fixed to guarantee that all data was in a usable format. For instance, non-numeric characters were removed from the monthly_rent column so that only numeric values remained and were converted to integers. Columns including size, rooms, bathroom, and parking were processed to extract numeric values from text-based entries, ensuring data type consistency for analysis. Additionally, temporal analyses were made possible by converting the completion_year field to integers.

We fixed the problems of missing data, inconsistent formatting, and type mismatches with these cleaning steps. As a result, the dataset was clear and trustworthy, offering a strong basis for precise analysis.

4.0 Data Visualization

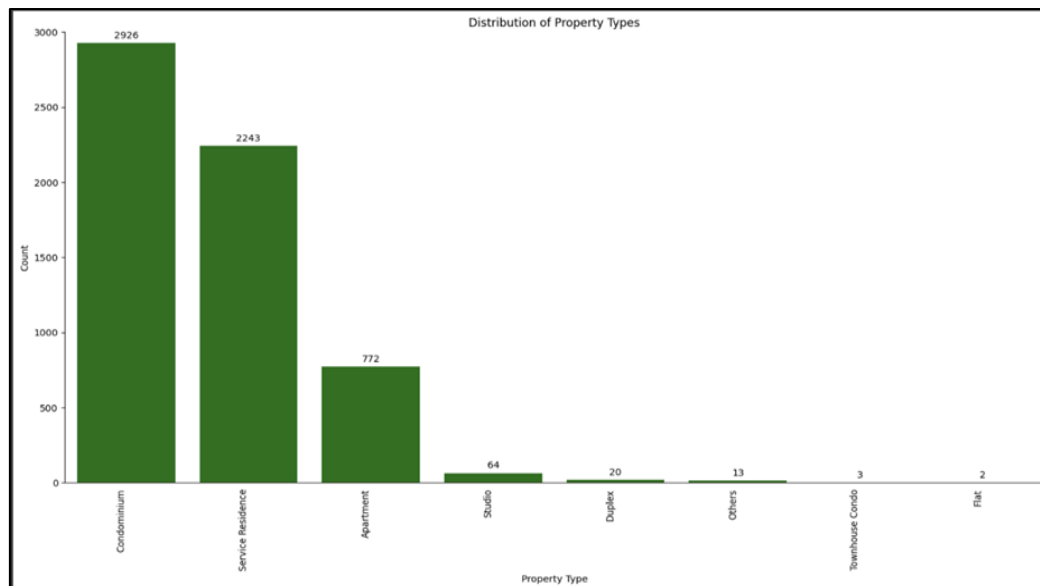


Figure 4.1 Distribution of Property Type

Figure 4.1 shows a bar graph for distribution of property type based on the cleaned dataset. From this figure, the highest property type is condominium with 2926 people are staying in condominium in Kuala Lumpur. The second highest and third highest are service residences and apartments with 2243 and 772 properties respectively. Other than that, Studio and duplex have 64 and 20 properties in Kuala Lumpur respectively. The second least and least distribution of properties are townhouse condominium and flat which have only 3 and 2 properties in Kuala Lumpur.

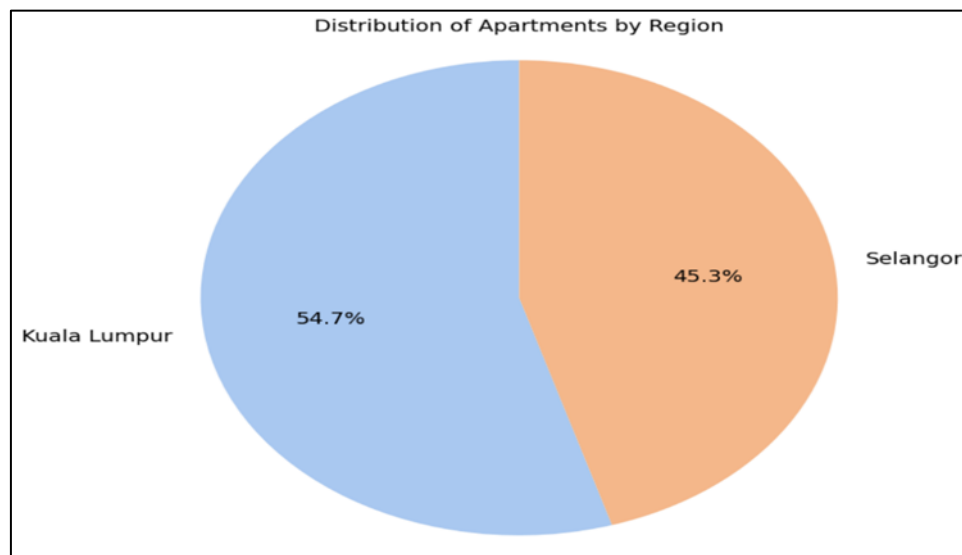


Figure 4.2 Distribution of Apartments by Region

Figure 4.2 shows the pie chart for distribution of Apartments by region for the cleaned dataset. The figure shows that Apartments in Kuala Lumpur is more than the Apartments in Selangor. The percentage of apartments in Kuala Lumpur is 54.7% and the percentage of apartments in Selangor is 45.3%.

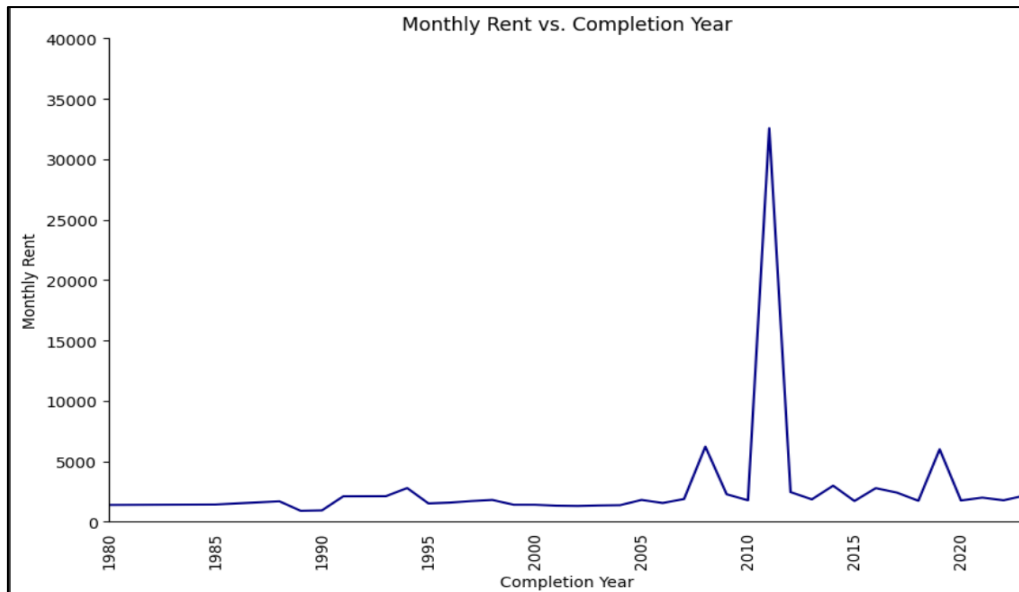


Figure 4.3 Monthly Rent vs. Completion Year

Figure 4.3 shows the relationship between monthly rent and completion year in line graph. The monthly rent was relatively low during the early era of 1980 to 1985. However, the monthly rent started to drop during 1988 till 1991 and rise again after that till 1995. After 1995, the monthly rent started to drop again and continue consistently until the year of 2004. The monthly rent rised again and continued to increase reached first peak and second peak in 2009 and 2011 respectively. However, the monthly rent drastically drop after 2012 and had frequent increase and decrease pattern and starts to rise slowly again in 2019. Though the monthly rent rise in 2019, it immediately drops at the end of 2019. The monthly rent become consistent starting 2020 until now.

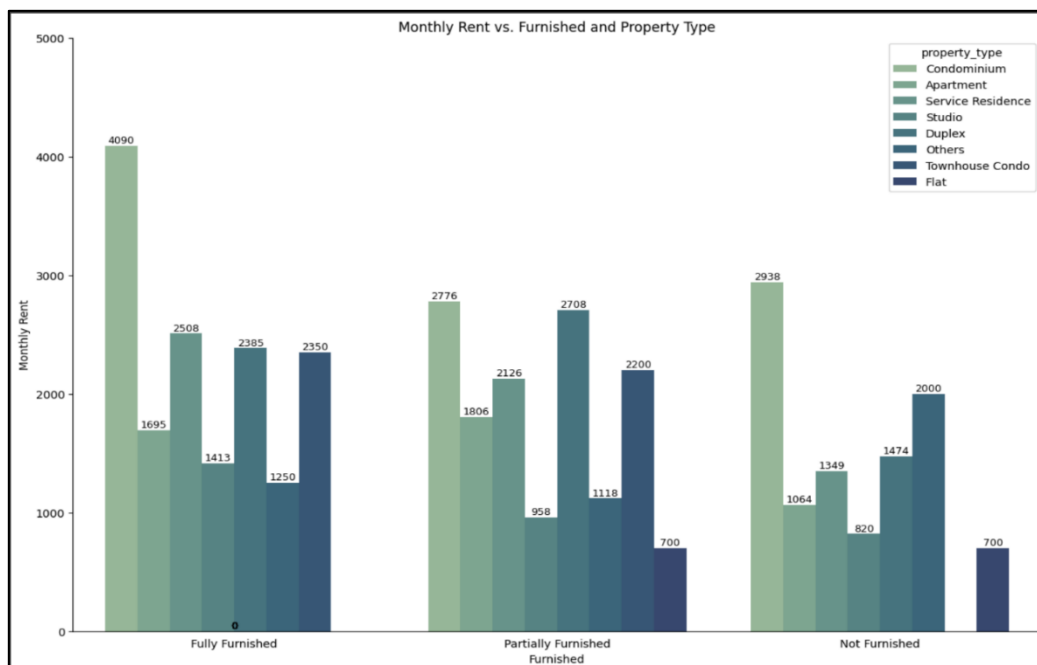


Figure 4.4 Monthly Rent vs. Furnished Property and Property Type.

Figure 4.4 shows the bar chart on 3 attributes which are monthly rent, furnished properties and property type. Condominium is the highest property type, which is fully furnished with monthly rent of RM4,090. The least property type in fully furnished properties category is townhouse condominium with monthly rent of RM1,250. In the partially furnished category, the highest rental property is

condominium with monthly rent RM2,776. The lowest rental price per month is RM700 which is flat for partially furnished property. For the not furnished category, the highest rental property is condominium with monthly rent of RM2,938. The lowest rental property is townhouse condominium which monthly rent is RM0.

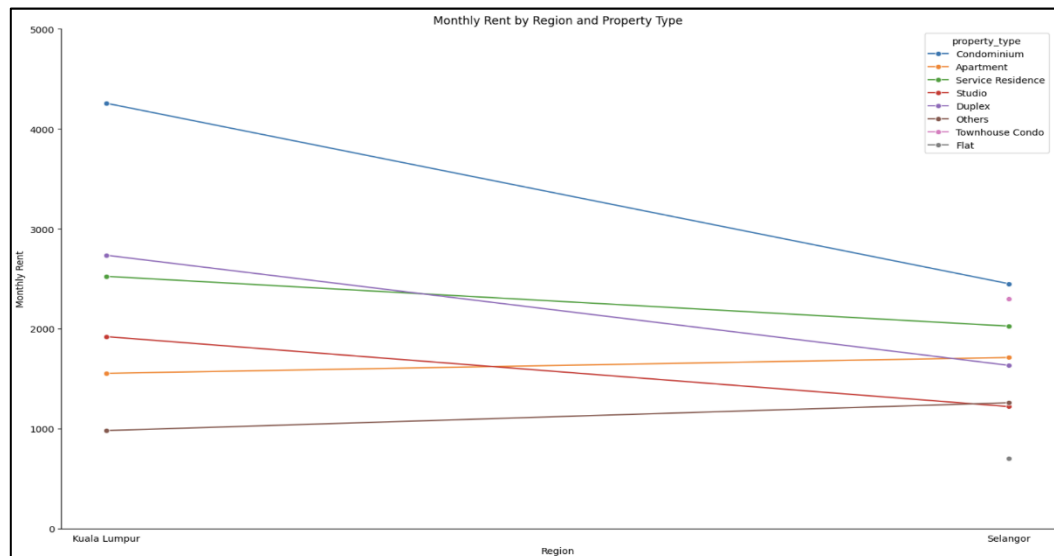


Figure 4.5 Monthly Rent by Region and Property Type

Figure 4.5 shows the line slope graph on the monthly rent by region and property type based on the cleaned dataset. The highest rental price property in Kuala Lumpur region is condominium whereas the lowest is other property types. The highest rental price property in Selangor region is condominium whereas the lowest is flat. The figure 3.5 also shows that flat and townhouse condominium is only available in Selangor region. Most of the property type have properties in Kuala Lumpur more than in Selangor. However, only apartments and other property types are mostly available more in Selangor than in Kuala Lumpur.

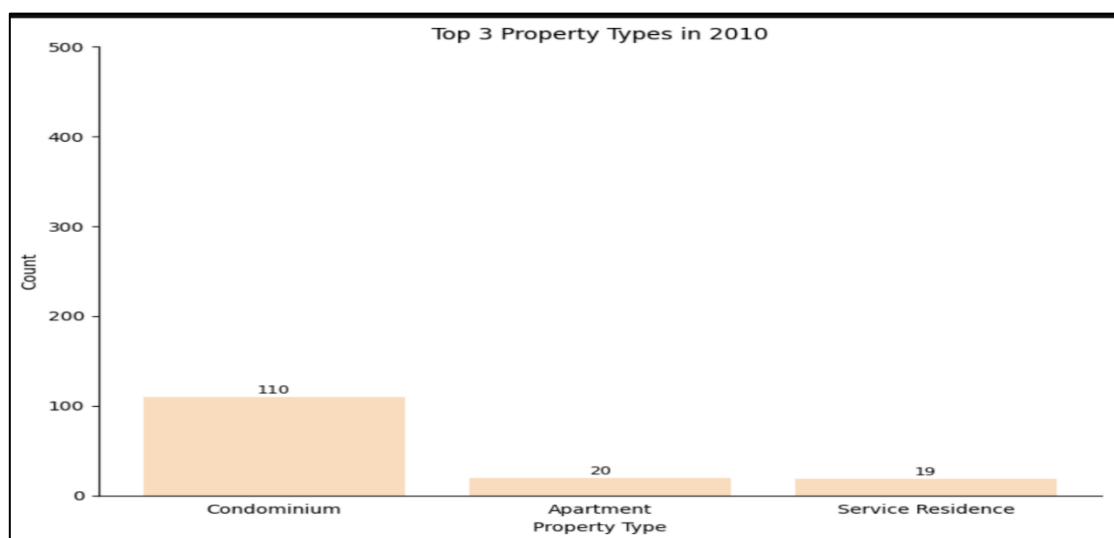


Figure 4.6.1 Top 3 Property Types in 2010

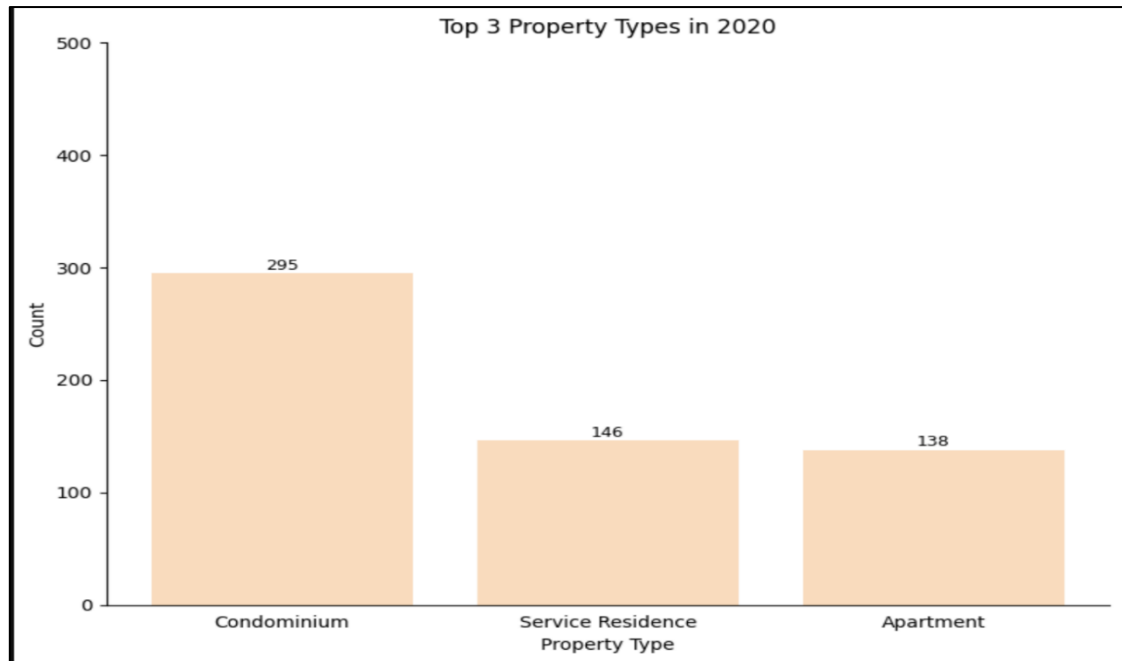


Figure 4.6.2 Top 3 Property Types in 2020

Figure 4.6.1 and 4.6.2 shows bar graph on the top 3 property type in 2010 and 2020 respectively based on the cleaned dataset. The top 1 property type bought by customers is condominium which rise from 110 unit in 2010 to 295 units in 2020. Apartment is the second highest property type in 2010 with 20 units and drops to third place in 2020 with 138 units. In 2010, Service residence was the third highest property sold with 19 units. However, it rise to the second highest property sold in 2020 with 146 units sold.

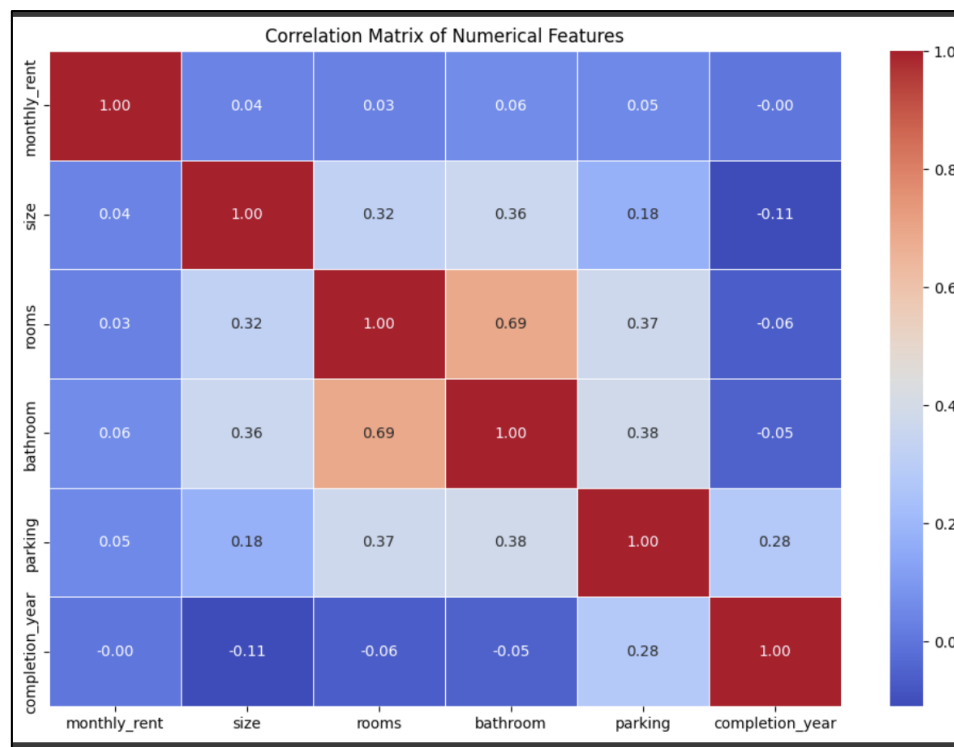


Figure 4.7 shows the correlation matrix heatmap of numerical features

Figure 4.7 shows the correlation between all the quantitative attributes. Number of rooms in the property and bathroom is the strong positive correlation with 0.69. This shows properties with more rooms tend

to have more bathrooms. There are few attributes with moderate positive correlation. Rooms and parking are moderate positive correlation with 0.37 because properties with more rooms usually have more parking spaces. Other than that, size and rooms are moderate positive correlation with 0.32 because larger properties have more rooms. Parking and completion year also have moderate positive correlation with 0.28 because recent properties have more parking spaces as people started to use more own vehicles. For completion year and monthly rent, there is no correlation between both attributes.

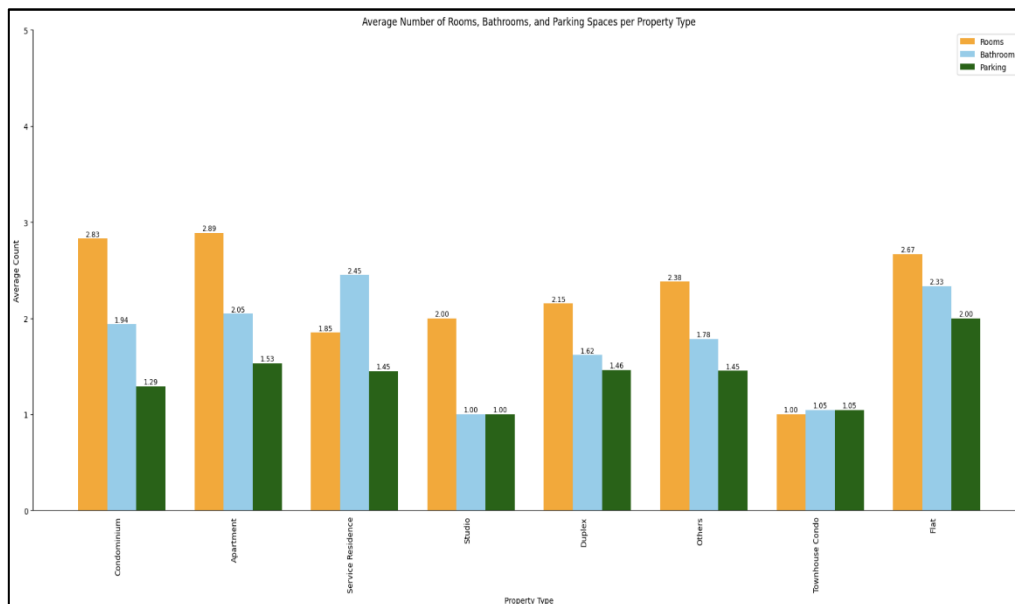


Figure 4.8 shows average number of rooms, bathrooms and parking spaces per property type

Figure 4.8 shows the grouped bar graph on the average number of rooms, bathrooms and parking spaces per property type based on the new dataset. From all the property type, apartment have the highest average number of rooms whereas service residence has the highest average number of bathrooms and flat has the highest average number of parking spaces. Meanwhile, the property with lowest average number of rooms is townhouse condominium. The property with lowest average number of bathrooms and parking spaces is studio.

5.0 Insights and Interpretation

Significant regional variations in rental prices are pointed out by the visualizations. Across property types, Kuala Lumpur and Selangor consistently command higher average rents, revealing higher demand, and potentially higher investment returns, for these regions. This likely reflects geographic differentials regarding location desirability, proximity to economic centers and amenities, among others, combined with living costs differentials. The relationship between completion year and monthly rent can be studied to show that the newer ones are more expensive. Although there have been some short-term variants on the trend, overall, the figure suggests that rental costs are increasing with time after 2010, and that properties from more recent years are more expensive to rent. The likely reason for this trend may be due to steadily increasing construction costs, inflation and new building amenities. This helps property owners strategically price the unit with regards to the year of construction.

In furnished properties, mainly apartments and condominiums, the rent is usually much higher than those of unfurnished or partly furnished units. In this light we can interpret these results to suggest that tenants value convenience and 'move in' readiness and are willing to pay a premium for furnished accommodation. This insight allows property owners to take advantage of it, by investing in furnishing it strategically, more especially when they are targeting young professionals and expats who will place a premium on convenience and hassle-free living. This gives the ability to increase rental price and

possibly to attract a more wanted tenant demographic, consequently increasing rental income and occupancy level.

By using the correlation matrix, we can begin to see how different property features interact and impact on rental prices. We find that while there is a weak positive correlation between the number of rooms and bathrooms with monthly rent, illustrating the fact that tenants prefer spaciousness and convenience, there is a moderate positive correlation between the number of parking spaces and monthly rent, indicating its value, particularly in cities with limited parking possibility. Most notably however, the monthly rent does not appear to have a strong negative correlation with property size as has been assumed and could be highly dependent on the type of property, or location. Preferring modern amenities which are popular in newer properties results in a weak positive correlation between completion year and rent. Similarly, older properties can also be modernized to include contemporary features which can improve their appeal and therefore justify higher rents for properties that complement changing expectations of the tenant.

An analysis of the distribution of property types within the rental market shows that “Apartments” and “Condominiums” account for a clearly dominant proportion of housing types, attesting to the high demand for these types of housing among tenants. Interestingly, from this view, we estimate reaching a market saturation level for these property types, though other types like “Service Apartments” and “Flats” enjoy a smaller share of the market. “Apartments” and “Condominiums” continue to be the staple, but those seeking an edge can also look at the less saturated markets like “Service Apartments” or “Flats” to meet niche occupier needs.

6.0 Conclusion

This report provides a comprehensive examination of rental pricing trends in the markets of Kuala Lumpur and Selangor through the use of data-driven insights and visualization tools. The monthly rental expenses are significantly influenced by the property type, furnishing condition, and geographical location, as indicated by significant results. The most prevalent property type across all furnishing categories was condominiums, as tenants' preference for move-in-ready properties that offer convenience and comfort is reflected in the attraction of completely furnished homes, which are the most desirable rentals. In response to the substantial demand for affordable and practicable housing options, apartment buildings and service homes constitute a substantial portion of the market.

The analysis of regional variations highlights that Kuala Lumpur commands slightly higher rents compared to Selangor, likely due to its economic significance, better infrastructure, and proximity to commercial hubs. In contrast, Selangor remains attractive for renters seeking more affordable options while still benefiting from accessibility to employment opportunities and urban amenities. This aligns with Malaysia's growing trend of urbanization, where renters are drawn to locations that balance cost and convenience.

There are numerous situational factors in Malaysia that elucidate tenants' decisions. University students and young professionals typically favour furnished flats and condominiums due to their advantageous locations near public transit centres, such as LRT/MRT stations. Tenants are compelled to prioritise properties with exceptional connectivity and adjacent amenities due to our country's escalating traffic congestion, thereby reducing the duration of their daily commutes. Expatriates and temporary labourers who require convenient accommodations for brief periods generate demand for fully furnished apartments.

The year of completion of a property and its rental prices are not significantly correlated. This indicates that rentals are typically higher in newer buildings due to their more modern amenities and infrastructure. Especially in cities where cost and space are significant factors, older homes that have been updated and are situated on outstanding lots are still selling well. Also, the study shows that features like parking spots are also highly valued, especially in city centres where the absence of parking enhances the appearance of residences.

The findings of the investigation offer a perceptive examination of the rental industry in Malaysia. This information can be used by tenants to identify reasonably priced homes that meet their needs, while landlords can optimise their rental strategy by concentrating on furnishing homes, selecting strategic locations, and accommodating shifting tenant preferences. These findings can be employed by policymakers to address housing affordability concerns, encourage improved urban design, and support initiatives that balance economic development with living standards.

In conclusion, this study bridges the knowledge gap in rental price determinants and sheds light on Malaysia's unique rental market dynamics. By comprehending tenant demands and property value determinants, stakeholders can make informed, data-driven decisions to excel in Malaysia's competitive real estate market.

7.0 References

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